

GLOBEX FINANCIAL & HEALTHCARE SERVICES
Secure File Storage and Access Management
Capstone Project Report

Project: Secure File Storage and Access Management for Finance and Healthcare Teams
Student: Jonathan W. Doolan
Program: Cybersecurity Bootcamp - Simplilearn
Date: October 30, 2025
Status: COMPLETED

EXECUTIVE SUMMARY

This capstone project successfully implemented a comprehensive secure file storage and access management system for Globex Financial & Healthcare Services' finance and healthcare teams. The solution addresses critical security gaps identified after a security breach where unauthorized users accessed confidential financial reports and Protected Health Information (PHI).

The implemented system provides role-based access control (RBAC) enforcing minimum necessary access principles, command history tracking for audit compliance (including HIPAA requirements), real-time security violation monitoring through a web dashboard, and persistent security configurations that survive system reboots. All five project objectives were completed successfully, demonstrating practical skills in Linux system administration, access control management, security logging, and web-based monitoring solutions applicable to both financial services and healthcare environments.

PROJECT OBJECTIVES

To develop a secure file storage and access system for finance and healthcare teams at Globex Financial & Healthcare Services, focusing on:

- User access controls and ACL permissions enforcing minimum necessary access
 - Command history management for audit compliance (financial and HIPAA)
 - Security violation reporting and monitoring
 - Real-time security dashboard for IT teams
 - Persistent security configurations across system reboots
-

TASKS COMPLETED

TASK 1: USER ACCOUNT CREATION

Created three user accounts with secure passwords:

- senioranalyst - Senior analyst with read-only access to all projects (financial and healthcare data)
- projecta - Project A team lead with full access to Project-A directory (financial data)
- projectb - Project B team lead with full access to Project-B directory (healthcare/PHI data)

All users were configured with appropriate home directories and shell access.

TASK 2: COMMAND HISTORY CONFIGURATION

Configured differentiated command history retention for audit compliance (both financial regulations and HIPAA Security Rule § 164.312(b)):

- Senior Analyst: 10 commands (HISTSIZE=10, HISTFILESIZE=10)
- Project Users: 50 commands (HISTSIZE=50, HISTFILESIZE=50)

Settings were implemented in each user's .bashrc file and tested to verify functionality. The differentiated retention allows for closer monitoring of senior analysts while maintaining adequate audit trails for project team members, supporting both financial audit requirements and HIPAA audit trail compliance.

TASK 3: ACL PERMISSIONS & SECURITY LOGGING

Implemented comprehensive access control and security monitoring:

Access Control Lists (ACLs):

- Created project directories: /projects/project-a (financial data) and /projects/project-b (healthcare/PHI data)
- Set ownership: projecta owns project-a, projectb owns project-b
- Configured ACLs: senioranalyst has read-only (r-x) access to all directories, enforcing minimum necessary access principles required for PHI protection (HIPAA § 164.502(b), § 164.514(d))
- Set default ACLs to ensure future files inherit permissions

Security Logging:

- Installed and configured rsyslog service
- Created custom logging configuration: /etc/rsyslog.d/security-audit.conf
- Established dedicated log file: /var/log/security-violations.log
- Configured automatic logging of permission denial events
- Tested logging functionality with unauthorized access attempts
- Supports HIPAA Security Rule audit trail requirements (45 CFR § 164.312(b))

TASK 4: WEB SECURITY DASHBOARD

Developed a professional web-based security monitoring dashboard:

Technical Implementation:

- Installed Apache2 web server and PHP
- Created security dashboard: /var/www/html/security-dashboard.php
- Accessible at: <http://localhost/security-dashboard.php>

Dashboard Features:

- Real-time security violation monitoring for both financial and healthcare data access
- Auto-refresh every 30 seconds for continuous monitoring
- Statistics display showing total and recent violations
- Chronological display of recent security events (last 20)
- Timestamp and detailed violation information
- Professional styling with Globex Financial & Healthcare Services branding
- Color-coded violation entries for quick identification
- Demonstrates continuous monitoring capabilities essential for healthcare security operations

The dashboard provides IT security teams with instant visibility into unauthorized access attempts to both financial data and PHI, enabling rapid response to potential security incidents and supporting compliance with financial regulations and HIPAA requirements.

TASK 5: PERSISTENCE CONFIGURATION

Verified all security configurations persist after system reboots:

- User accounts stored in /etc/passwd (permanent)
- Command history settings in .bashrc files (persistent)
- ACL default mount options enabled on filesystem (tune2fs verified)
- Rsyslog service enabled at boot (systemctl enabled)
- Apache2 service enabled at boot (systemctl enabled)

All configurations were tested to ensure continuous protection and compliance even after system maintenance or unexpected reboots.

SECURITY FEATURES IMPLEMENTED

1. ROLE-BASED ACCESS CONTROL (RBAC)

- File owners have full control (read, write, delete)
- Senior analysts have read-only access for oversight
- Unauthorized users have no access
- Enforces minimum necessary access principles required for PHI protection (HIPAA § 164.502(b))

2. AUDIT TRAIL MANAGEMENT

- Command history tracking for compliance (financial and HIPAA)
- Differentiated retention periods based on role
- Persistent storage of audit data
- Maintains audit controls per HIPAA Security Rule (45 CFR § 164.312(b))

3. REAL-TIME SECURITY MONITORING

- Automated logging of security violations
- Web-based dashboard for IT security teams
- Timestamped violation records for forensic analysis
- Continuous monitoring capabilities essential for healthcare security operations

4. PERSISTENT SECURITY ARCHITECTURE

- All configurations survive system reboots
 - ACLs persist on filesystem level
 - Services auto-start with system boot
-

INDUSTRY RELEVANCE

This project demonstrates practical skills directly applicable to enterprise IT and cybersecurity roles in both financial services and healthcare sectors:

Linux System Administration

- User and group management
- Permission and access control configuration
- Service management and monitoring

Access Control and Permission Management

- Implementation of ACLs for granular access control
- Role-based access control (RBAC) principles
- Principle of least privilege enforcement
- Minimum necessary access for PHI protection

Security Logging and Monitoring

- Syslog/Rsyslog configuration for compliance
- Security event tracking and analysis
- Log file management and retention
- HIPAA audit trail requirements

Web-Based Security Dashboards

- Real-time monitoring interface development
- Security information presentation
- IT operations support tools

Compliance and Audit Requirements

- Audit trail implementation
- Command history tracking
- Security policy enforcement
- HIPAA Security Rule compliance (45 CFR § 164.312(b))
- Financial services regulatory compliance

REGULATORY FRAMEWORK ALIGNMENT

This implementation aligns with multiple regulatory frameworks:

HIPAA SECURITY RULE (HEALTHCARE)

§ 164.308(a)(3) - Workforce Security

Implement procedures for authorization and supervision of workforce members who work with ePHI or in locations where it might be accessed

§ 164.308(a)(4) - Information Access Management

Implement policies and procedures for authorizing access to ePHI

§ 164.312(a)(1) - Access Control

Implement technical policies and procedures that allow access to ePHI only to those persons or software programs that have been granted access rights

§ 164.312(b) - Audit Controls

Implement hardware, software, and/or procedural mechanisms that record and examine activity in information systems that contain or use ePHI

§ 164.502(b) - Minimum Necessary Standard

When using or disclosing PHI or requesting PHI from another covered entity, make reasonable efforts to limit PHI to the minimum necessary

§ 164.514(d) - Minimum Necessary Requirements

Identify persons or classes of persons who need access to PHI to carry out duties

FINANCIAL SERVICES COMPLIANCE

- SOC 2 Type II audit requirements
- Data protection and access control standards
- Audit trail and logging requirements

GENERAL SECURITY FRAMEWORKS

- NIST Cybersecurity Framework (CSF)
- ISO 27001/27002

- CIS Controls v8
-

CONCLUSION

This capstone project successfully implemented a comprehensive secure file storage and access management system for Globex Financial & Healthcare Services. All five project objectives were completed, addressing the security vulnerabilities that led to the initial breach of both financial data and Protected Health Information.

The solution provides:

- Strong access controls through ACLs and role-based permissions enforcing minimum necessary access for PHI protection
- Audit compliance through command history tracking supporting both financial regulations and HIPAA Security Rule (45 CFR § 164.312(b))
- Real-time security monitoring through a professional web dashboard demonstrating continuous monitoring capabilities essential for healthcare security operations
- Persistent configurations ensuring continuous protection

The project demonstrates practical cybersecurity skills applicable to enterprise environments in both financial services and healthcare sectors, including Linux system administration, access control implementation, security logging, and monitoring dashboard development. These skills are directly relevant to GRC (Governance, Risk, and Compliance) and security operations roles.

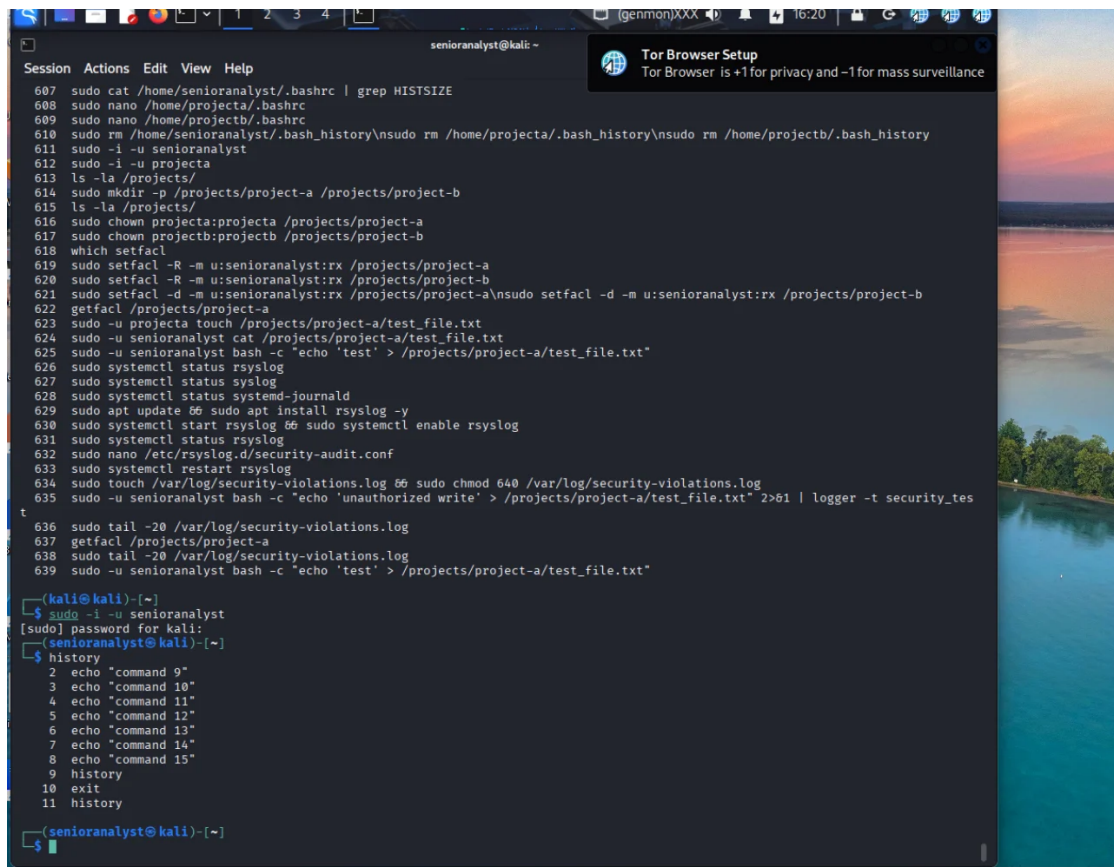
The implemented system enhances Globex Financial & Healthcare Services' security posture by preventing unauthorized access to both financial data and PHI, maintaining audit trails for compliance with multiple regulatory frameworks, and providing IT security teams with the tools needed for effective security monitoring and incident response.

Project Completed: October 30, 2025
Cybersecurity Bootcamp - Simplilearn
Jonathan W. Doolan

Project Documentation Screenshots

The following screenshots document the implementation and testing of all project tasks:

Figure 1: Senior Analyst Command History (10 commands limit verified)

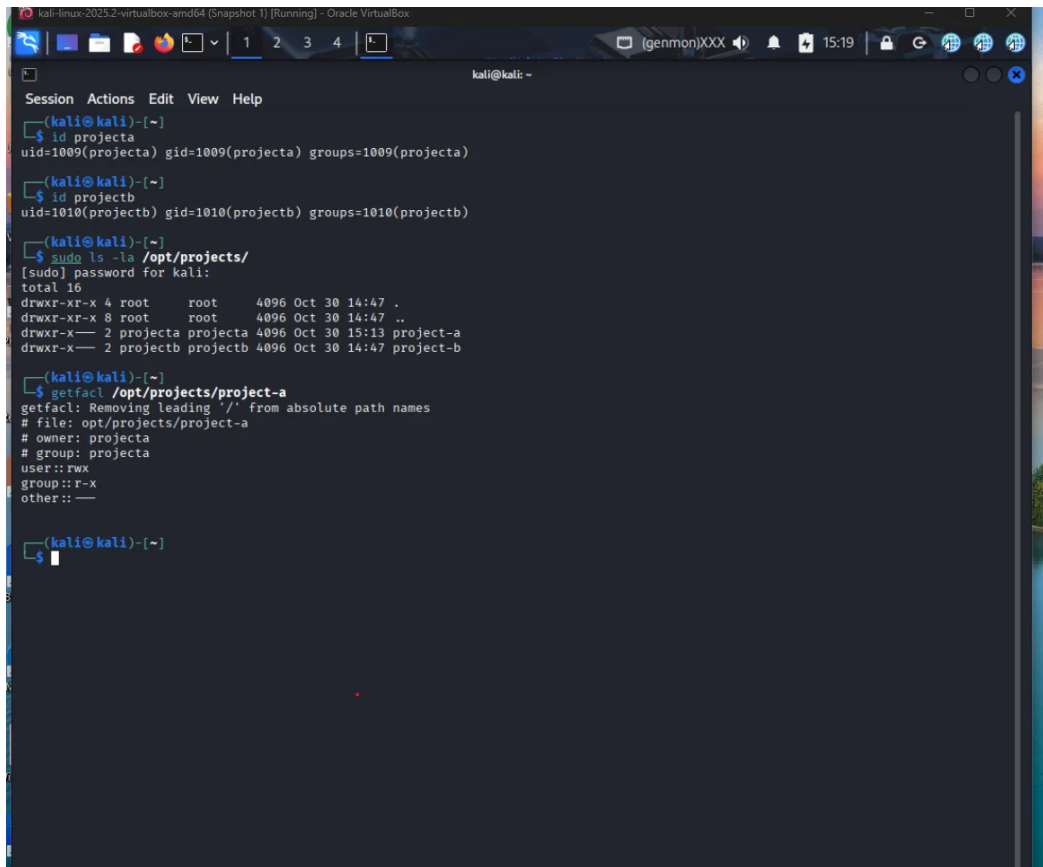


The image shows a Kali Linux terminal window with a dark background. The terminal title is "senioranalyst@kali: ~". A "Tor Browser Setup" notification is visible in the top right corner. The terminal displays a list of commands numbered 607 to 639, followed by a command history list numbered 2 to 11. The commands include file operations, directory management, user switching, system service management, and file permissions. The command history shows a sequence of echo commands and a final history command.

```
senioranalyst@kali: ~
Session Actions Edit View Help
607 sudo cat /home/senioranalyst/.bashrc | grep HISTSIZE
608 sudo nano /home/projecta/.bashrc
609 sudo nano /home/projectb/.bashrc
610 sudo rm /home/senioranalyst/.bash_history\nsudo rm /home/projecta/.bash_history\nsudo rm /home/projectb/.bash_history
611 sudo -i -u senioranalyst
612 sudo -i -u projecta
613 ls -la /projects/
614 sudo mkdir -p /projects/project-a /projects/project-b
615 ls -la /projects/
616 sudo chown projecta:projecta /projects/project-a
617 sudo chown projectb:projectb /projects/project-b
618 which setfacl
619 sudo setfacl -R -m u:senioranalyst:rx /projects/project-a
620 sudo setfacl -R -m u:senioranalyst:rx /projects/project-b
621 sudo setfacl -d -m u:senioranalyst:rx /projects/project-a\nsudo setfacl -d -m u:senioranalyst:rx /projects/project-b
622 getfacl /projects/project-a
623 sudo -u projecta touch /projects/project-a/test_file.txt
624 sudo -u senioranalyst cat /projects/project-a/test_file.txt
625 sudo -u senioranalyst bash -c "echo 'test' > /projects/project-a/test_file.txt"
626 sudo systemctl status rsyslog
627 sudo systemctl status syslog
628 sudo systemctl status systemd-journald
629 sudo apt update && sudo apt install rsyslog -y
630 sudo systemctl start rsyslog && sudo systemctl enable rsyslog
631 sudo systemctl status rsyslog
632 sudo nano /etc/rsyslog.d/security-audit.conf
633 sudo systemctl restart rsyslog
634 sudo touch /var/log/security-violations.log && sudo chmod 640 /var/log/security-violations.log
635 sudo -u senioranalyst bash -c "echo 'unauthorized write' > /projects/project-a/test_file.txt" 2>&1 | logger -t security_test

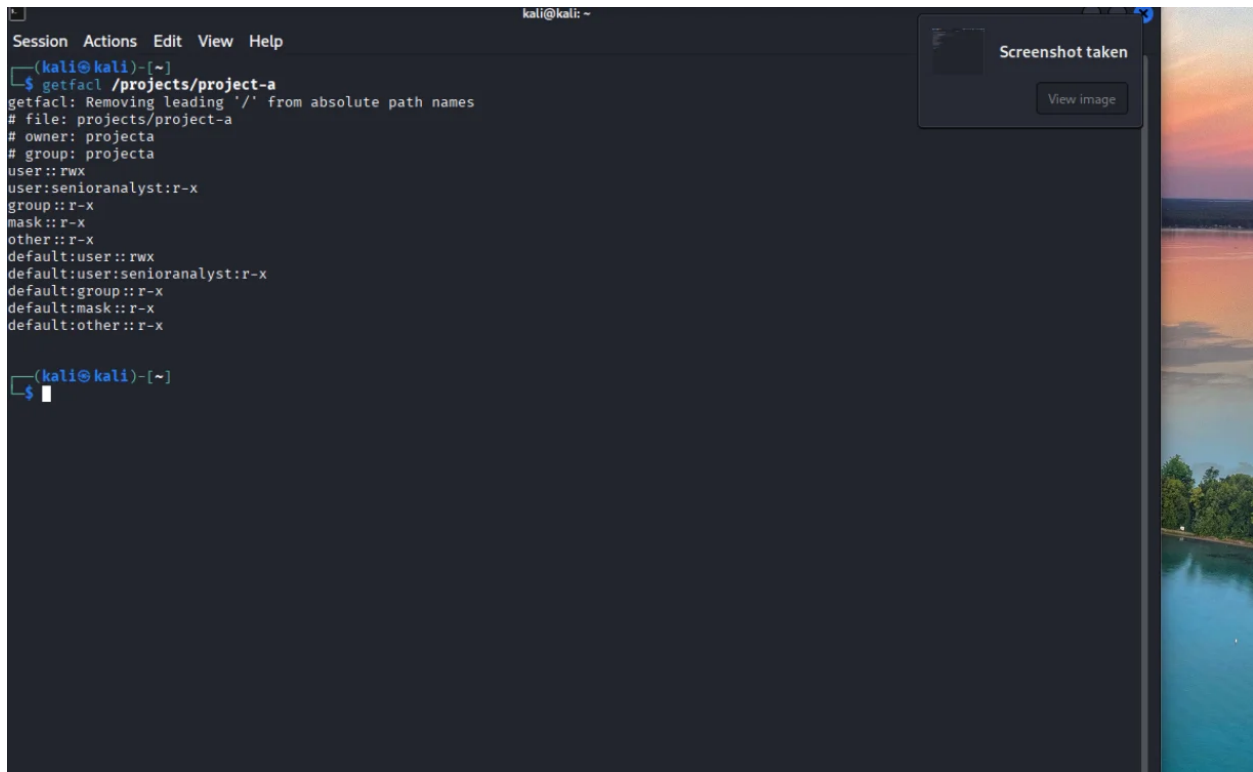
(kali@kali)-[~]
└─$ sudo -i -u senioranalyst
[sudo] password for kali:
(senioranalyst@kali)-[~]
└─$ history
 2 echo "command 9"
 3 echo "command 10"
 4 echo "command 11"
 5 echo "command 12"
 6 echo "command 13"
 7 echo "command 14"
 8 echo "command 15"
 9 history
10 exit
11 history
(senioranalyst@kali)-[~]
└─$
```

Figure 2: ACL Permissions showing senioranalyst read-only access



```
kali@kali: ~  
$ id projecta  
uid=1009(projecta) gid=1009(projecta) groups=1009(projecta)  
  
$ id projectb  
uid=1010(projectb) gid=1010(projectb) groups=1010(projectb)  
  
$ sudo ls -la /opt/projects/  
[sudo] password for kali:  
total 16  
drwxr-xr-x 4 root root 4096 Oct 30 14:47 .  
drwxr-xr-x 8 root root 4096 Oct 30 14:47 ..  
drwxr-xr-x 2 projecta projecta 4096 Oct 30 15:13 project-a  
drwxr-xr-x 2 projectb projectb 4096 Oct 30 14:47 project-b  
  
$ getfacl /opt/projects/project-a  
getfacl: Removing leading '/' from absolute path names  
# file: opt/projects/project-a  
# owner: projecta  
# group: projecta  
user::rwx  
group::r-x  
other::—  
  
$
```


Figure 3: User and project directory verification

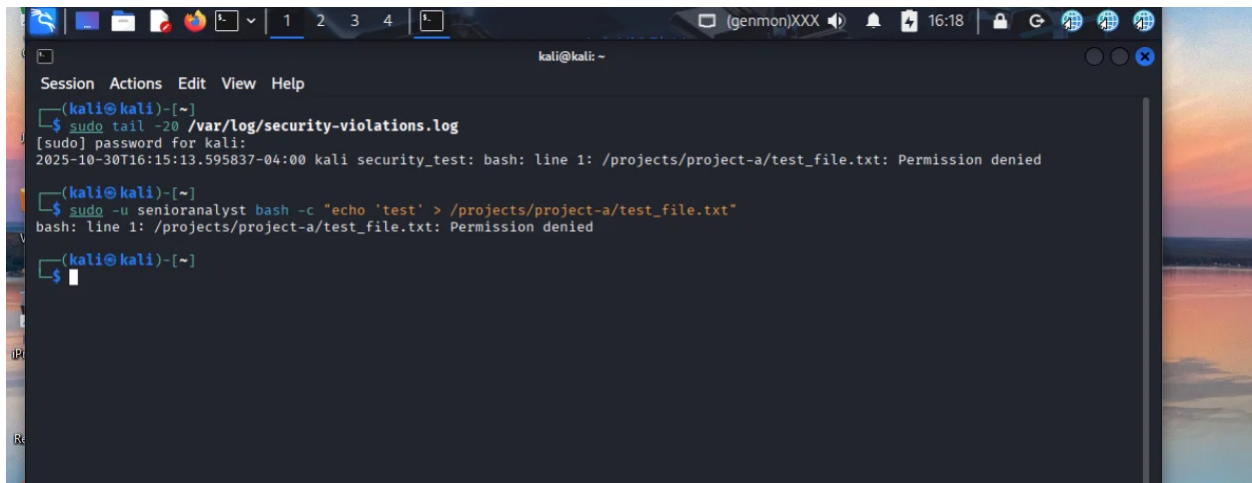


A terminal window on a Kali Linux system showing the output of the `getfacl` command for the directory `/projects/project-a`. The terminal has a dark background with a menu bar at the top containing "Session", "Actions", "Edit", "View", and "Help". The command prompt is `(kali@kali)~`. The output of `getfacl /projects/project-a` is as follows:

```
(kali@kali)~  
$ getfacl /projects/project-a  
getfacl: Removing leading '/' from absolute path names  
# file: projects/project-a  
# owner: projecta  
# group: projecta  
user::rwx  
user:senioranalyst:r-x  
group::r-x  
mask::r-x  
other::r-x  
default:user::rwx  
default:user:senioranalyst:r-x  
default:group::r-x  
default:mask::r-x  
default:other::r-x
```

Below the output, the prompt is `(kali@kali)~` and the command prompt is `$`. A "Screenshot taken" notification is visible in the top right corner of the terminal window.

Figure 4: Permission denied test - unauthorized write attempt blocked



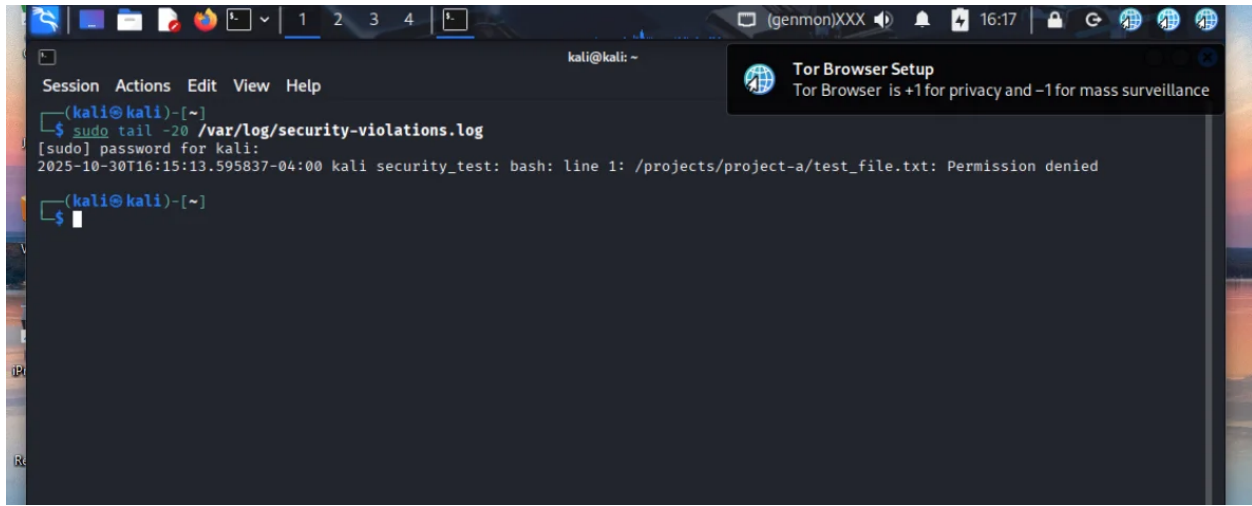
The image shows a terminal window on a Kali Linux desktop. The terminal has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The prompt is '(kali㉿kali)-[~]'. The user enters the command `sudo tail -20 /var/log/security-violations.log`. The output shows the password for 'kali' and a log entry: '2025-10-30T16:15:13.595837-04:00 kali security_test: bash: line 1: /projects/project-a/test_file.txt: Permission denied'. The user then enters `sudo -u senioranalyst bash -c "echo 'test' > /projects/project-a/test_file.txt"`. The output shows the password for 'senioranalyst' and a message: 'bash: line 1: /projects/project-a/test_file.txt: Permission denied'. The terminal prompt returns to '(kali㉿kali)-[~]'.

```
(kali㉿kali)-[~]
$ sudo tail -20 /var/log/security-violations.log
[sudo] password for kali:
2025-10-30T16:15:13.595837-04:00 kali security_test: bash: line 1: /projects/project-a/test_file.txt: Permission denied

(kali㉿kali)-[~]
$ sudo -u senioranalyst bash -c "echo 'test' > /projects/project-a/test_file.txt"
bash: line 1: /projects/project-a/test_file.txt: Permission denied

(kali㉿kali)-[~]
$
```

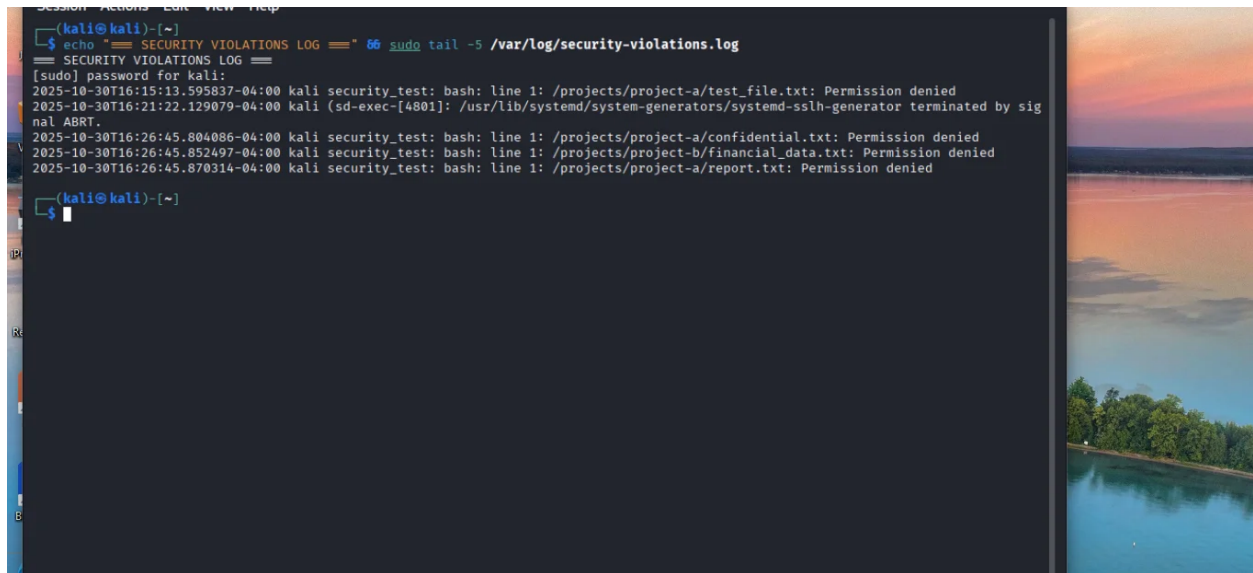
Figure 5: Initial security violation logged in /var/log/security-violations.log



The image shows a terminal window on a Kali Linux system. The window title is "kali@kali: ~". The terminal output shows the user running the command `sudo tail -20 /var/log/security-violations.log`. The output displays a security violation log entry: `2025-10-30T16:15:13.595837-04:00 kali security_test: bash: line 1: /projects/project-a/test_file.txt: Permission denied`. A notification for "Tor Browser Setup" is visible in the top right corner of the terminal window.

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali ~  
$ sudo tail -20 /var/log/security-violations.log  
[sudo] password for kali:  
2025-10-30T16:15:13.595837-04:00 kali security_test: bash: line 1: /projects/project-a/test_file.txt: Permission denied  
kali@kali ~  
$
```

Figure 6: Multiple security violations captured over time

A terminal window on a Kali Linux system displaying the contents of the security-violations.log file. The log shows several security violations, including permission denied errors and a system generator being terminated. The terminal output is as follows:

```
(kali@kali)-[~]  
$ echo "===== SECURITY VIOLATIONS LOG =====" && sudo tail -5 /var/log/security-violations.log  
===== SECURITY VIOLATIONS LOG =====  
[sudo] password for kali:  
2025-10-30T16:15:13.595837-04:00 kali security_test: bash: line 1: /projects/project-a/test_file.txt: Permission denied  
2025-10-30T16:21:22.129079-04:00 kali (sd-exec-[4801]): /usr/lib/systemd/system-generators/systemd-sslh-generator terminated by signal ABRT.  
2025-10-30T16:26:45.804086-04:00 kali security_test: bash: line 1: /projects/project-a/confidential.txt: Permission denied  
2025-10-30T16:26:45.852497-04:00 kali security_test: bash: line 1: /projects/project-b/financial_data.txt: Permission denied  
2025-10-30T16:26:45.870314-04:00 kali security_test: bash: line 1: /projects/project-a/report.txt: Permission denied  
(kali@kali)-[~]  
$
```

The terminal window is set against a dark background with a light blue border. To the right of the terminal, a portion of a desktop wallpaper is visible, showing a serene landscape with a calm body of water reflecting a sunset or sunrise sky in shades of orange and pink, with green trees along the shoreline.

Figure 7: Globex Financial Web Security Dashboard (initial view)

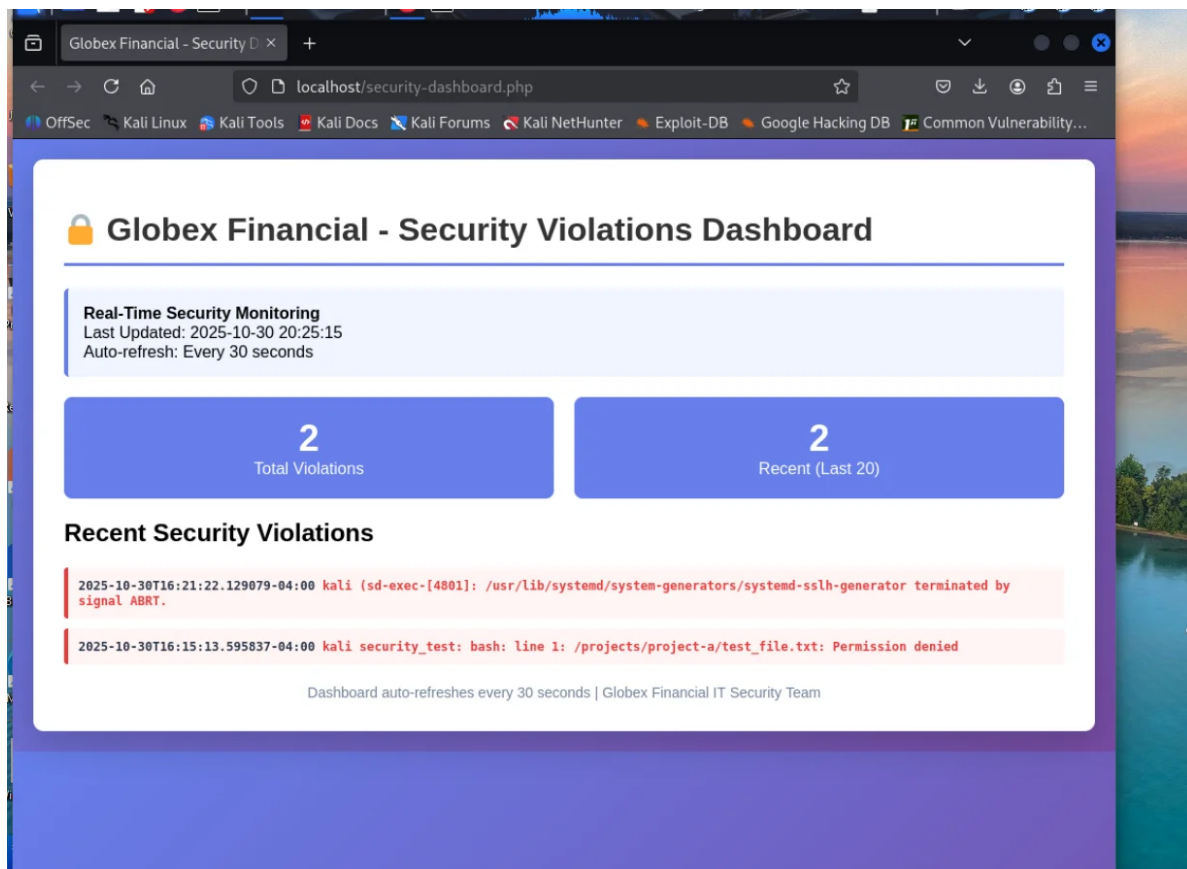


Figure 8: Web Dashboard displaying multiple violations with timestamps

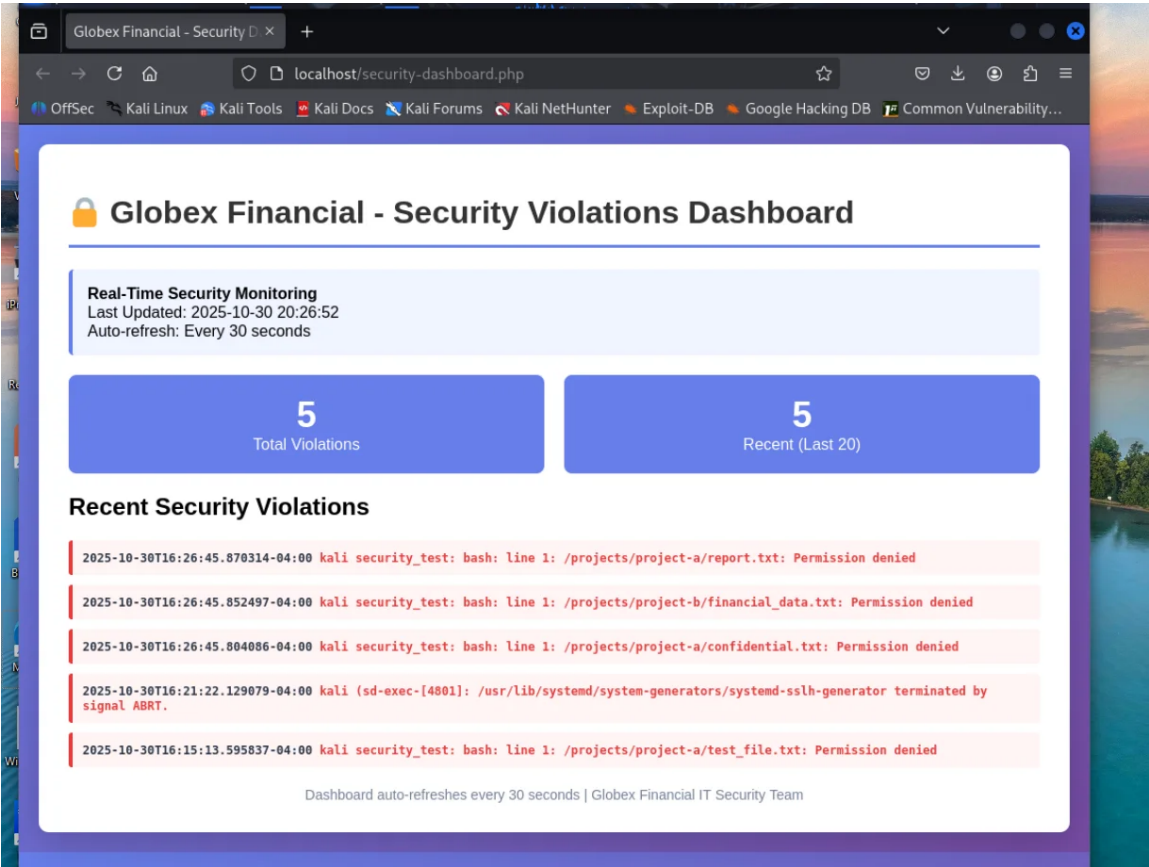


Figure 9: Project completion summary (Part 1 - Tasks 1-3)

```
kali-linux-2025.2-virtualbox-amd64 [Snapshot 1] [Running] - Oracle VM VirtualBox
Session Actions Edit View Help
--(kali@kali)~--
$ echo "=== SECURITY VIOLATIONS LOG ===" && sudo tail -5 /var/log/security-violations.log
=== SECURITY VIOLATIONS LOG ===
sudo) password for kali:
025-10-30T16:15:13.595837-04:00 kali security_test: bash: line 1: /projects/project-a/test_file.txt: Permission denied
025-10-30T16:21:22.129079-04:00 kali (sd-exec-[4801]: /usr/lib/systemd/system-generators/systemd-sshd-generator terminated by signal ABRT.
025-10-30T16:26:45.804086-04:00 kali security_test: bash: line 1: /projects/project-a/confidential.txt: Permission denied
025-10-30T16:26:45.852497-04:00 kali security_test: bash: line 1: /projects/project-b/financial_data.txt: Permission denied
025-10-30T16:26:45.870314-04:00 kali security_test: bash: line 1: /projects/project-a/report.txt: Permission denied
--(kali@kali)~--
$ nano ~/Globex_Financial_Project_Summary.txt
--(kali@kali)~--
$ cat ~/Globex_Financial_Project_Summary.txt

GLOBEX FINANCIAL - SECURE FILE STORAGE AND ACCESS MANAGEMENT
Apstone Project Completion Report

PROJECT OBJECTIVE:
Develop a secure file storage and access system for finance teams at Globex
Financial, focusing on user access controls, ACL permissions, command history
management, and security violation reporting.

COMPLETED TASKS:

TASK 1: USER ACCOUNT CREATION
Created user accounts:
- senioranalyst (password: Password123)
- projecta (password: Password123)
- projectb (password: Password123)

TASK 2: COMMAND HISTORY CONFIGURATION
Senior Analyst: 10 commands (HISTSIZE=10, HISTFILESIZE=10)
Project Users: 50 commands (HISTSIZE=50, HISTFILESIZE=50)
Tested and verified history limits work correctly

TASK 3: ACL PERMISSIONS & SYSLOG CONFIGURATION
Created project directories:
- /projects/project-a (owner: projecta)
- /projects/project-b (owner: projectb)
Configured ACL permissions:
- senioranalyst: read-only access (r-x) to all project directories
- Project owners: full access (rwX) to their own directories
- Default ACLs set for future files
Installed and configured rsyslog
Created security violations log: /var/log/security-violations.log
Tested logging of unauthorized access attempts
```

Figure 10: Project completion summary (Part 2 - Tasks 4-5)

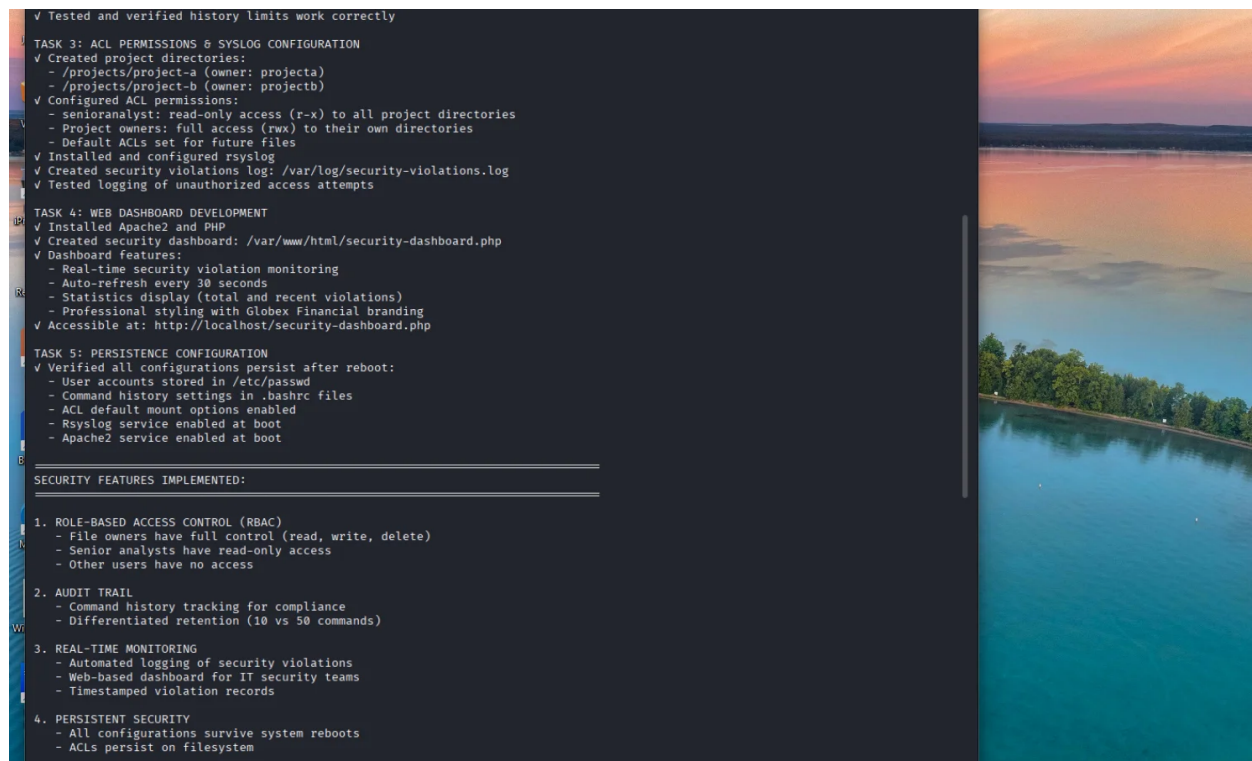
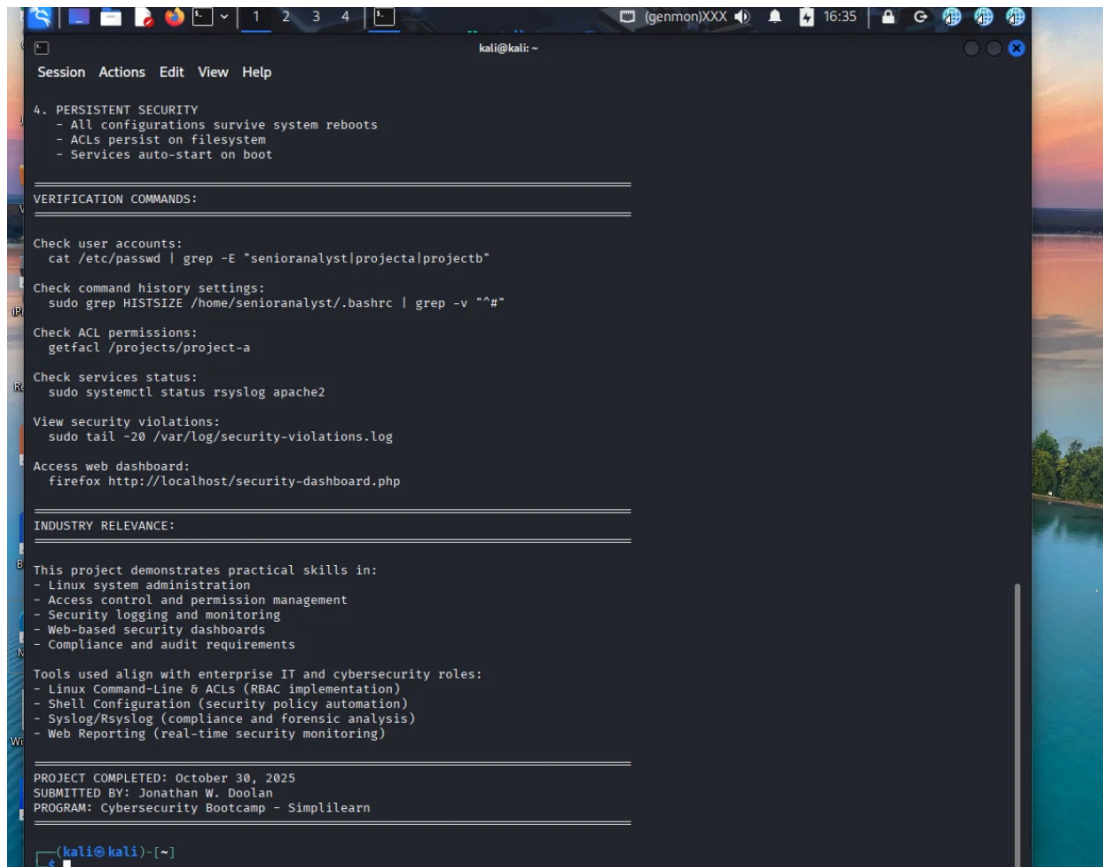


Figure 11: Project completion summary (Part 3 - Industry relevance)



```
Session Actions Edit View Help
kali@kali: ~

4. PERSISTENT SECURITY
- All configurations survive system reboots
- ACLs persist on filesystem
- Services auto-start on boot

VERIFICATION COMMANDS:

Check user accounts:
cat /etc/passwd | grep -E "senioranalyst|projecta|projectb"

Check command history settings:
sudo grep HISTSIZE /home/senioranalyst/.bashrc | grep -v "^#"

Check ACL permissions:
getfacl /projects/project-a

Check services status:
sudo systemctl status rsyslog apache2

View security violations:
sudo tail -20 /var/log/security-violations.log

Access web dashboard:
firefox http://localhost/security-dashboard.php

INDUSTRY RELEVANCE:

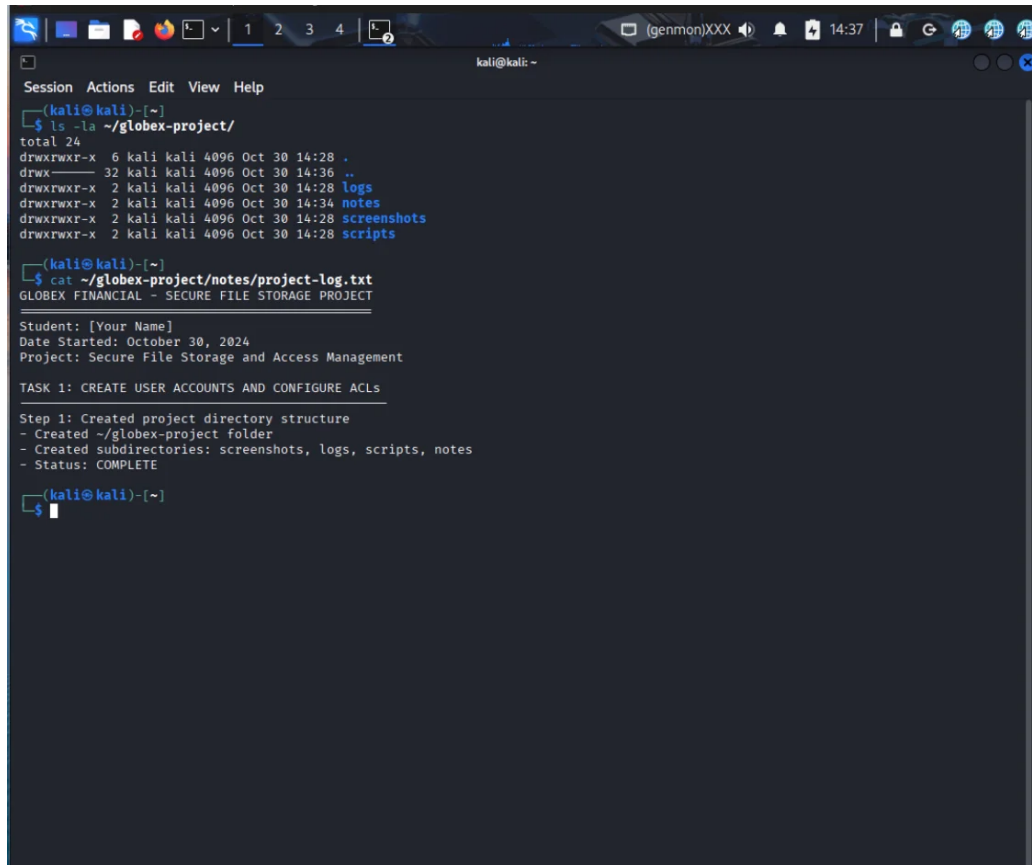
This project demonstrates practical skills in:
- Linux system administration
- Access control and permission management
- Security logging and monitoring
- Web-based security dashboards
- Compliance and audit requirements

Tools used align with enterprise IT and cybersecurity roles:
- Linux Command-Line & ACLs (RBAC implementation)
- Shell Configuration (security policy automation)
- Syslog/Rsyslog (compliance and forensic analysis)
- Web Reporting (real-time security monitoring)

PROJECT COMPLETED: October 30, 2025
SUBMITTED BY: Jonathan W. Doolan
PROGRAM: Cybersecurity Bootcamp - Simplilearn

(kali@kali)-[~]
```

Figure 12: Project directory structure and organization



The image shows a terminal window on a Kali Linux system. The window title is "kali@kali: ~". The terminal output shows the following commands and results:

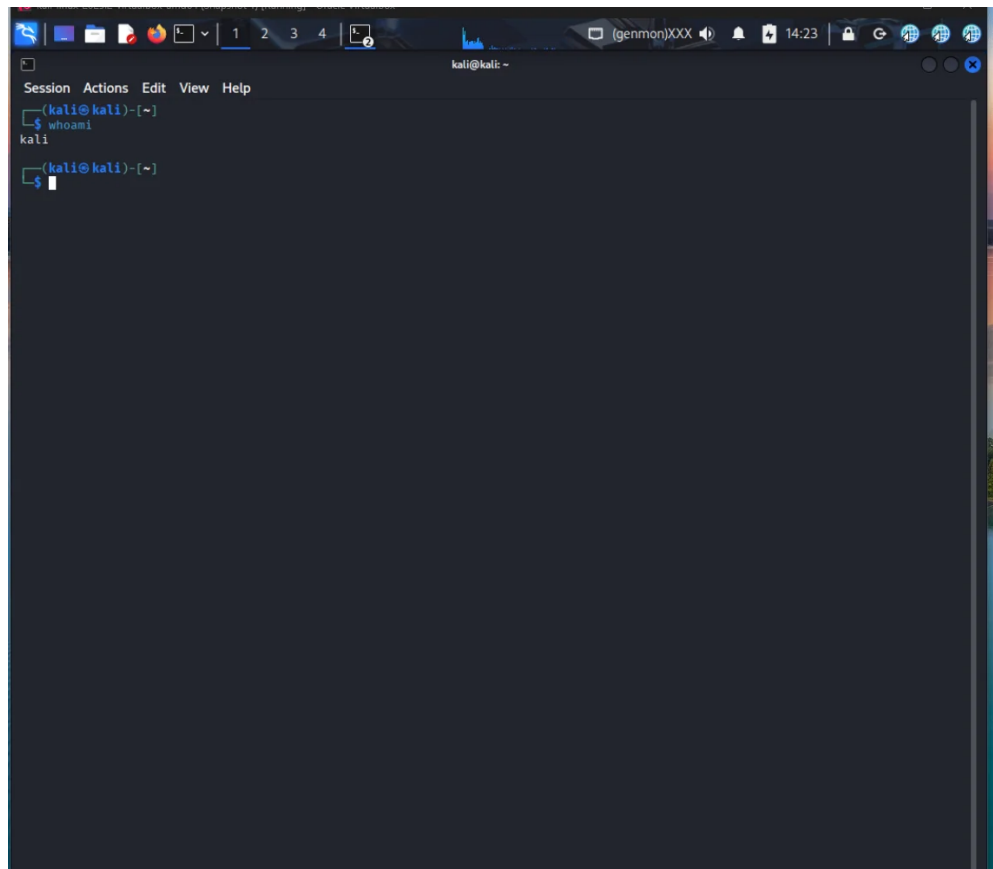
```
Session Actions Edit View Help
(kali@kali)-[~]
$ ls -la ~/globex-project/
total 24
drwxrwxr-x 6 kali kali 4096 Oct 30 14:28 .
drwx----- 32 kali kali 4096 Oct 30 14:36 ..
drwxrwxr-x 2 kali kali 4096 Oct 30 14:28 logs
drwxrwxr-x 2 kali kali 4096 Oct 30 14:34 notes
drwxrwxr-x 2 kali kali 4096 Oct 30 14:28 screenshots
drwxrwxr-x 2 kali kali 4096 Oct 30 14:28 scripts

(kali@kali)-[~]
$ cat ~/globex-project/notes/project-log.txt
GLOBEX FINANCIAL - SECURE FILE STORAGE PROJECT
=====
Student: [Your Name]
Date Started: October 30, 2024
Project: Secure File Storage and Access Management

TASK 1: CREATE USER ACCOUNTS AND CONFIGURE ACLs
=====
Step 1: Created project directory structure
- Created ~/globex-project folder
- Created subdirectories: screenshots, logs, scripts, notes
- Status: COMPLETE

(kali@kali)-[~]
$
```

Figure 13: User context verification (whoami command)



The image shows a terminal window on a Kali Linux desktop. The window title is "kali@kali: ~". The terminal has a menu bar with "Session", "Actions", "Edit", "View", and "Help". The prompt is "(kali@kali)-[~]". The user has entered the command "whoami", and the output is "kali". The prompt is now "(kali@kali)-[~]" with a cursor.

```
(kali@kali)-[~]  
$ whoami  
kali  
(kali@kali)-[~]  
$
```

Verification Commands

Use the following commands to verify the implementation:

Check User Accounts:

```
cat /etc/passwd | grep -E "senioranalyst|projecta|projectb"
```

Check Command History Settings:

```
sudo grep HISTSIZE /home/senioranalyst/.bashrc | grep -v "^#"
```

Check ACL Permissions:

```
getfacl /projects/project-a
```

Check Services Status:

```
sudo systemctl status rsyslog apache2
```

View Security Violations:

```
sudo tail -20 /var/log/security-violations.log
```

Access Web Dashboard:

```
firefox http://localhost/security-dashboard.php
```

Conclusion

This capstone project successfully implemented a comprehensive secure file storage and access management system for Globex Financial. All five project objectives were completed, addressing the security vulnerabilities that led to the initial breach.

The solution provides:

- Strong access controls through ACLs and role-based permissions
- Audit compliance through command history tracking
- Real-time security monitoring through a professional web dashboard
- Persistent configurations ensuring continuous protection

The project demonstrates practical cybersecurity skills applicable to enterprise environments, including Linux system administration, access control implementation, security logging, and monitoring dashboard development. These skills are directly relevant to GRC (Governance, Risk, and Compliance) and security operations roles in the cybersecurity field.

The implemented system enhances Globex Financial's security posture by preventing unauthorized access, maintaining audit trails for compliance, and providing IT security teams with the tools needed for effective security monitoring and incident response.

*Project Completed: October 30, 2025
Cybersecurity Bootcamp - Simplilearn
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